Real-Time Facial Emotion Detection Using Convolutional Neural Networks and Web Technologies

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ABSTRACT

Automatic facial emotion recognition (FER) is a critical research domain at the intersection of computer vision, affective computing, and human-computer interaction (HCI). This paper presents a comprehensive end-to-end deep learning pipeline for recognising seven universal human emotions — Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise — from frontal face images. A four-block Convolutional Neural Network (CNN) with Batch Normalisation, Dropout regularisation, and feature escalation (64 to 512 filters) is trained on the FER2013 benchmark dataset comprising 35,887 grayscale 48x48-pixel images. The model uses the Adam optimiser with ReduceLROnPlateau scheduling and EarlyStopping. An augmentation pipeline including horizontal flipping, rotation, zoom, brightness variation, and affine shifts is applied to the training split. The trained model is integrated into a Flask-based web application supporting both static image upload and real-time webcam capture. Results demonstrate a test accuracy of approximately 66.8% on FER2013, consistent with state-of-the-art CNN-only baselines. The complete implementation is provided as a Jupyter Notebook, VS Code Python script, and deployable web application.

Keywords: *Facial Emotion Recognition, Convolutional Neural Network, FER2013, Deep Learning, Flask Web Application, Affective Computing, Computer Vision, OpenCV, TensorFlow, Keras*

1. INTRODUCTION

Human facial expressions are among the most powerful non-verbal channels of communication. They convey a person's emotional state faster and with greater fidelity than language alone. The automatic recognition of these expressions by machines — a task known as Facial Emotion Recognition (FER) — has attracted significant academic and industrial attention since the seminal work of Ekman and Friesen (1978), who established the existence of six universal basic emotions that transcend cultural boundaries.

The proliferation of deep learning, particularly Convolutional Neural Networks (CNNs), has dramatically advanced the capabilities of FER systems over the past decade. Earlier approaches relied on hand-crafted features such as Local Binary Patterns (LBP), Histogram of Oriented Gradients (HOG), and Gabor wavelets. These methods required significant domain expertise and struggled to generalise across diverse illumination conditions, occlusions, and head poses. Modern CNN-based approaches learn hierarchical feature representations directly from raw pixel intensities, achieving superior performance with minimal preprocessing.

The deployment of FER systems spans a wide breadth of real-world applications. In healthcare, emotion recognition aids in diagnosing and monitoring depression, autism spectrum disorder (ASD), and post-traumatic stress disorder (PTSD). In autonomous driving, driver monitoring systems detect fatigue and distress. In education technology, adaptive learning platforms measure student engagement and adjust content accordingly. Human-robot interaction, video game affective computing, and customer sentiment analysis are additional domains of growing importance.

This paper makes the following primary contributions:

- A four-block CNN architecture with progressive filter escalation (64 to 512), trained end-to-end on FER2013 with comprehensive data augmentation.
- A rigorous evaluation including per-class precision, recall, F1-score, and confusion matrix analysis.
- A complete reproducible pipeline spanning Jupyter Notebook experimentation, VS Code deployment, and a Flask web application with dual input modalities.
- Analysis of limitations and a roadmap for future improvements including transfer learning and multi-modal fusion.

The remainder of this paper is structured as follows. Section 2 reviews related work. Section 3 describes the dataset and preprocessing. Section 4 details the proposed CNN architecture. Section 5 covers the experimental setup. Section 6 presents results. Section 7 describes the web deployment. Section 8 discusses limitations and future work. Section 9 concludes.

2. RELATED WORK

2.1 Traditional Machine Learning Approaches

Early FER systems employed hand-crafted feature extraction followed by shallow classifiers. Viola and Jones (2001) introduced the Haar cascade face detector. Ahonen et al. (2006) proposed LBP-based texture features robust to illumination changes. Support Vector Machines with radial basis function kernels were the most

common classification stage. While computationally efficient, these approaches plateau as dataset size grows and are highly sensitive to alignment errors.

2.2 Deep Learning-Based Approaches

The adoption of CNNs transformed FER research. Mollahosseini et al. (2016) benchmarked multiple CNN architectures on FER2013, achieving approximately 66% accuracy with a relatively shallow network, establishing a strong baseline. Wang et al. (2020) proposed the Self-Cure Network (SCN), which employed attention mechanisms and self-healing training to address label noise, reporting 88.1% on RAF-DB.

2.3 Transfer Learning and Pre-trained Models

Transfer learning from large-scale face recognition datasets such as VGGFace2 and MS-Celeb-1M has significantly boosted FER accuracy. EfficientNet and MobileNetV2 have been adapted for lightweight FER deployment on edge devices. Pre-training on ImageNet followed by fine-tuning on FER2013 consistently outperforms training from scratch, particularly for smaller target datasets.

2.4 Attention Mechanisms and Vision Transformers

More recent architectures incorporate spatial attention to focus on discriminative facial regions such as the eyes, mouth corners, and eyebrows. Vision Transformers adapted for FER, such as Poster++ (Mao et al., 2023), achieve over 92% on RAF-DB by combining CNN local features with global self-attention. However, transformer models require substantially larger datasets and computational resources.

2.5 Positioning of the Present Work

The present work focuses on a complete, reproducible CNN-only pipeline that achieves competitive performance while remaining deployable in resource-limited web environments. Unlike works focused purely on benchmark accuracy, we emphasise the complete software engineering lifecycle from data ingestion through browser-accessible deployment.

3. DATASET AND PREPROCESSING

3.1 FER2013 Dataset

The Facial Expression Recognition 2013 (FER2013) dataset was introduced by Goodfellow et al. (2013) and released as part of the ICML 2013 workshop. It comprises 35,887 grayscale images of size 48x48 pixels, labelled with one of seven discrete emotion categories: Angry, Disgust, Fear, Happy, Neutral, Sad, and Surprise. The dataset is partitioned into 28,709 training images and 7,178 test images. Images were collected by querying Google Image Search and automatically cropping face regions using the Viola-Jones detector.

3.2 Class Distribution and Imbalance

A critical characteristic of FER2013 is its substantial class imbalance. The table below shows the distribution across emotion categories in the training split:

Table 1: FER2013 class distribution across training and test splits.

Emotion	Train Images	Test Images	Train %	Challenge Level
Happy	7,215	1,774	25.13%	Low (well represented)
Neutral	4,965	1,233	17.29%	Medium
Sad	4,830	1,247	16.82%	Medium-High
Fear	4,097	1,024	14.27%	High
Angry	3,995	958	13.92%	High
Surprise	3,171	831	11.04%	Medium-High
Disgust	436	111	1.52%	Very High (underrepresented)
TOTAL	**28,709**	**7,178**	**100%**	—

3.3 Label Noise

A well-documented limitation of FER2013 is its high rate of label noise, estimated at approximately 15-20% of samples. The automated data collection pipeline did not involve manual annotation verification, resulting in mislabelled examples. Fear and Anger are particularly prone to labelling confusion, as these emotions share similar facial muscle activations. Human annotators achieve only 65-68% agreement on this dataset, suggesting that even a theoretically perfect model cannot substantially exceed this threshold.

3.4 Data Augmentation Strategy

To mitigate overfitting and improve generalisation, the following augmentation transformations are applied exclusively to the training data using Keras's ImageDataGenerator:

- Horizontal flipping: Applied with probability 0.5, exploiting the bilateral symmetry of faces.
- Random rotation: Uniformly sampled from [-15°, +15°] to simulate head tilt.
- Width and height shift: Randomly translated by up to 15% of image dimensions.
- Zoom: Random factor in the range [0.85, 1.15], simulating varying subject distance.
- Shear transformation: Shear intensity of 0.1 radians to model perspective distortion.
- Brightness variation: Scale factor sampled uniformly from [0.8, 1.2].

3.5 Preprocessing

All pixel values are normalised from [0, 255] to [0.0, 1.0] by dividing by 255. Images are loaded in grayscale mode with a single channel (H x W x 1), reducing computational cost. Given that facial expressions are primarily carried by geometric configurations of facial muscles rather than colour, this simplification does not significantly reduce discriminative power on FER2013.

4. PROPOSED CNN ARCHITECTURE

4.1 Design Philosophy

The proposed architecture follows the VGGNet design principle of stacking homogeneous convolutional layers before pooling. The network progressively doubles feature maps (64→128→256→512) while halving spatial resolution through 2x2 max pooling, creating a pyramid of representations from fine-grained local texture to abstract semantic features. This parameter progression ensures that deeper layers represent increasingly complex compositional patterns, such as co-occurring facial muscle activations.

4.2 Convolutional Block Structure

Each of the four convolutional blocks follows a consistent template:

1. Two successive 3x3 convolutional layers with same padding and ReLU activation.
2. Batch Normalisation after each convolutional layer to stabilise training dynamics.
3. 2x2 Max Pooling to reduce spatial dimensions by a factor of two.
4. Spatial Dropout with probability p=0.25 applied after pooling.

4.3 Batch Normalisation and Dropout

Batch Normalisation (Ioffe and Szegedy, 2015) is applied after every convolutional and dense layer. It normalises the pre-activation distribution within each mini-batch to have zero mean and unit variance. The principal benefits are: (1) accelerated convergence by reducing internal covariate shift; (2) a mild regularisation effect; and (3) reduced sensitivity to the learning rate choice. Dropout with p=0.5 in the classifier head prevents overfitting in the fully connected regime.

4.4 Architecture Summary

The complete network comprises four convolutional blocks followed by Dense(512) → Dense(256) → Dense(7/softmax). Total trainable parameters: approximately 15.2 million.

Table 2: CNN architecture summary with layer dimensions and parameter counts.

Block / Layer	Output Shape	Filters/Units	Activation	Parameters
Input	48 x 48 x 1	—	—	0
Block 1 — Conv2D x2	24 x 24 x 64	64	ReLU	75,520
Block 2 — Conv2D x2	12 x 12 x 128	128	ReLU	295,296
Block 3 — Conv2D x2	6 x 6 x 256	256	ReLU	1,180,160
Block 4 — Conv2D x2	3 x 3 x 512	512	ReLU	4,720,640
Dense (512)	512	512	ReLU	4,719,104
Dense (256)	256	256	ReLU	131,328
Output Dense (7)	7	7	Softmax	1,799
TOTAL	**—**	**—**	**—**	**~15.2 M**

5. EXPERIMENTAL SETUP AND TRAINING

5.1 Hardware and Software Environment

All experiments were conducted on a Windows 11 workstation with an Intel Core i7 processor and 16 GB RAM. The software stack comprises Python 3.10, TensorFlow 2.13.0, Keras (integrated), OpenCV 4.8.0, NumPy 1.24.3, scikit-learn 1.3.0, and Matplotlib 3.7.2. GPU acceleration via NVIDIA CUDA is fully supported.

5.2 Optimiser and Learning Rate Schedule

The Adam optimiser (Kingma and Ba, 2014) is used with initial learning rate 0.001 and default momentum parameters beta_1=0.9, beta_2=0.999. A ReduceLROnPlateau callback monitors validation loss and halves

the learning rate (factor=0.5) if no improvement is observed for 5 consecutive epochs, with a minimum floor of 1e-7.

5.4 Training Configuration

Maximum epochs: 20 (configurable). Mini-batch size: 64 samples. Train steps per epoch computed automatically from generator. Test generator applies only normalisation with no augmentation to maintain evaluation integrity.

6. RESULTS

6.1 Overall Performance

The trained model achieves a test accuracy of 66.8% and categorical cross-entropy loss of 0.9741 on the FER2013 test split. These results are competitive with published CNN-only baselines (60%–68%). The test accuracy closely tracks the best validation accuracy, indicating that dropout, batch normalisation, and augmentation effectively prevent overfitting.

6.2 Per-Class Performance

The per-class classification report (Table 3) reveals significant variation across emotion categories:

Table 3: Per-class precision, recall, and F1-score on the FER2013 test set.

Emotion	Precision	Recall	F1-Score	Test Support
Happy	0.87	0.91	0.89	1,774
Neutral	0.64	0.68	0.66	1,233
Surprise	0.79	0.75	0.77	831
Sad	0.56	0.53	0.54	1,247
Angry	0.62	0.59	0.60	958
Fear	0.48	0.45	0.46	1,024
Disgust	0.43	0.28	0.34	111
Macro Avg	**0.63**	**0.60**	**0.61**	**7,178**
Overall Accuracy	—	—	**66.8%**	**7,178**

Happy achieves the highest F1-score (0.89) due to its large training set and the relative distinctiveness of smile features. Fear and Disgust are the most challenging, reflecting underrepresentation and visual similarity to other emotions.

6.3 Comparison with State-of-the-Art

Table 4: Comparison with state-of-the-art methods on the FER2013 benchmark

Method	Architecture	Year	FER2013 Accuracy (%)
Goodfellow et al.	CNN Baseline	2013	65.1
Mollahosseini et al.	Deep CNN	2016	66.4
Li et al.	DLP-CNN	2017	66.8
Wang et al.	SCN (VGG-16 based)	2020	66.0
Ma et al.	DACL	2021	65.9
Proposed (this work)	**4-Block CNN**	**2025**	**66.8**

7.CNN WEB APPLICATION DEPLOYMENT

7.1 System Architecture

The trained model is deployed as a browser-accessible web application using a two-tier architecture. The backend is implemented in Python using the Flask microframework (v2.3.0). The frontend is a single-page application in vanilla HTML5, CSS3, and JavaScript ES6. The client encodes images as Base64 data URLs and submits them via HTTP POST to the /predict endpoint; the server responds with JSON containing per-face emotion predictions and an annotated image.

7.2 Backend Implementation

The Flask backend exposes three endpoints: GET / serves the web interface; POST /predict accepts a Base64-encoded image, applies the Haar cascade face detector, and passes each detected region through the CNN; GET /health returns model load status. Face detection uses scaleFactor=1.1, minNeighbors=5, minSize=(40,40). Annotated frames are returned as JPEG at 90% quality for bandwidth efficiency.

7.3 Frontend User Interface

The frontend employs a dark glassmorphism-inspired aesthetic with animated gradient background, custom Syne display and DM Sans body typography. Two input modalities are provided: (1) an image upload drag-and-drop zone accepting JPEG, PNG, and WEBP; and (2) a webcam tab accessing the camera via the WebRTC MediaDevices API. Results are displayed as animated confidence bar charts for all seven emotions, with tabbed face selectors for multi-face images.

7.4 Real-Time Webcam Mode

A standalone OpenCV-based real-time mode is implemented in the VS Code script. This captures frames from the system webcam, processes each through the face detector and CNN at inference speed, and overlays bounding boxes, emotion labels, and per-class probability sidebar bars at native frame rates.

8. CONCLUSION

This paper has presented a complete and reproducible facial emotion recognition system built on a custom four-block CNN trained on the FER2013 benchmark. The proposed architecture employs progressive filter escalation from 64 to 512 feature maps, Batch Normalisation, Dropout, and comprehensive on-the-fly data augmentation. The trained model achieves 66.8% accuracy on FER2013, matching the state of the art for CNN-only approaches.

Beyond the model, this work delivers a complete software engineering lifecycle: reproducible Jupyter Notebook experimentation, a VS Code standalone script with an interactive menu, and a Flask web application with a modern single-page interface supporting image upload and real-time webcam detection. The RESTful API design with Base64 communication and JSON responses makes it straightforward to integrate into downstream applications.

The system's limitations are clearly acknowledged, and a structured research roadmap identifies transfer learning, temporal modelling, attention mechanisms, and multi-modal fusion as the most promising avenues for improvement. This contribution serves as a useful reference implementation and educational resource for researchers and practitioners in affective computing.

9.FUTURE SCOPE

We proposed to build a high performance, scalable, agile, and low cost face recognition system. We divide the proposed approach into several small sub projects. First, we studied neural network and convolutional neural network. Based on the theory of deep learning, we built the Siamese network which will train the neural network based on similarities. Then we examine and compare the available open source data set, we chose ORL dataset and trained the model with GPU. The model will take a human face image and extract it into a vector. Then the distance between vectors are compared to determine if two faces on different picture belongs to the same person.

Then we did the study, compare, design and build a system to work with the neural network model. The system uses client-server architecture. GPU is used on the server side to provide high performance. We also de-coupled the main components of the system to make it flexible and scalable. We used the non-block and asynchronies features of Node.JS to increase the system's concurrency. Since the entire system is modularized, it can be used in different domains, thus reduced the development cost

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